

- Q.27 What do you mean analog transducer and digital transducer? (CO2)
- Q.28 Write a short note on Piezoelectric transducer. (CO3)
- Q.29 Explain the working principle of well type Manometer. (CO4)
- Q.30 Explain any one method for speed measurement. (CO4)
- Q.31 Explain any two display devices used in the instrumentation system. (CO1)
- Q.32 List applications of thermocouple gauge. (CO4)
- Q.33 Draw and explain the block diagram of signal conditioning system. (CO1)
- Q.34 Write a short note on vibration measurement. (CO4)
- Q.35 Explain moisture and atmospheric humidity. (CO4)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Give the working of (a) Dc tachometer (b) photoelectric speed sensor. (CO4)
- Q.37 Explain pH measurement. Give its advantages, disadvantages and applications. (CO4)
- Q.38 Explain the construction, working and limitations of LVDT with diagram. (CO4)

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5th Sem / Elect, Eltx. Sub : Instrumentation

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 _____ is an active transducer. (CO1)
- a) Thermocouple b) Piezoelectric
c) None of these d) All of these
- Q.2 A quartz crystal generates voltage proportional to _____ applied upon it. (CO3)
- a) Stress b) Displacement
c) Torque d) All of these
- Q.3 LVDTs have _____ hysteresis. (CO3)
- a) High b) Low
c) Infinite d) Zero
- Q.4 Capacitive transducer is used to measure _____. (CO3)
- a) Displacement b) Thickness
c) Level d) All of these
- Q.5 Temperature is measured by _____. (CO4)
- a) Barometer b) Manometer
c) Thermistor d) Hygrometer

- Q.6 Air is a mixture of _____ (CO4)
 a) Dry air b) Water vapor
 c) Both option A & B d) None of the above
- Q.7 Force is defined as (CO4)
 a) Velocity x displacement
 b) Mass x displacement
 c) Velocity x acceleration
 d) mass x acceleration
- Q.8 Piezoelectric effect is used in _____ (CO3)
 a) Microphones b) Micropumps
 c) Inkjet printer d) All of the above
- Q.9 Potentiometer is a _____ transducer (CO4)
 a) Displacement b) Force
 c) Torque d) None of these
- Q.10 Pressure is measured in _____ (CO4)
 a) Newton b) Pascal
 c) Weber d) Tesla

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Humidity is measured in Pascals. (True/ False) (CO4)

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- Q.12 The force acting on a body which produces rotation is known as _____ (CO4)
- Q.13 Manometers are used to measure _____. (CO4)
- Q.14 What are analog transducers? (CO1)
- Q.15 What is measurement? (CO1)
- Q.16 Name any positive temperature coefficient material? (Co4)
- Q.17 Give applications of load cell. (CO3)
- Q.18 Name any data presentation element. (CO1)
- Q.19 What is a primary sensing element? (CO1)
- Q.20 On which phenomenon bourdon tube works?(CO4)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain the working principle of PVDT. (CO3)
- Q.22 Explain any five factors affecting the choice of a transducer. (CO1)
- Q.23 Write a short note on Level measurement. (CO4)
- Q.24 State the applications of thermopile. (CO4)
- Q.25 List the advantages and disadvantages of electromagnetic flow meter. (CO4)
- Q.26 Explain the working of strain gauge load cell. (CO4)

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